

Logistic Population Growth



The equation for logistic growth in discrete time

LEARNING OBJECTIVES

The equation for logistic growth in discrete time

The definition of density-dependent growth

The equation for logistic growth in discrete time

The definition of density-dependent growth

Basic properties of the model

The equation for logistic growth in discrete time

The definition of density-dependent growth

Basic properties of the model

Strange behavior of the (discrete time) model, such as damped oscillations and chaos

Geometric growth

$$N_{t+1} = N_t + N_t r$$

Geometric growth

$$N_{t+1} = N_t + N_t r$$

Logistic growth

$$N_{t+1} = N_t + N_t r_{max} \left(1 - \frac{N_t}{K} \right)$$

where

- r_{max} is the growth rate when N_t is close to 0.
- K is the carrying capacity

Logistic growth is an example of **density-dependent growth**

Logistic growth is an example of **density-dependent growth**

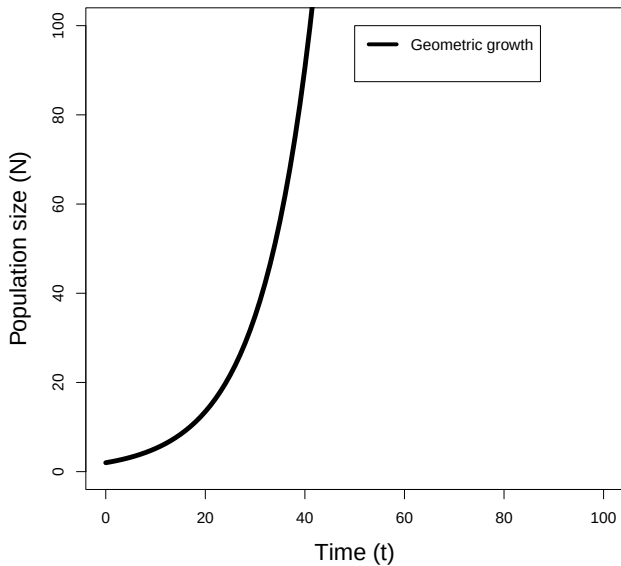
Definition: Population growth rate *is* affected by population size (N).

Logistic growth is an example of **density-dependent growth**

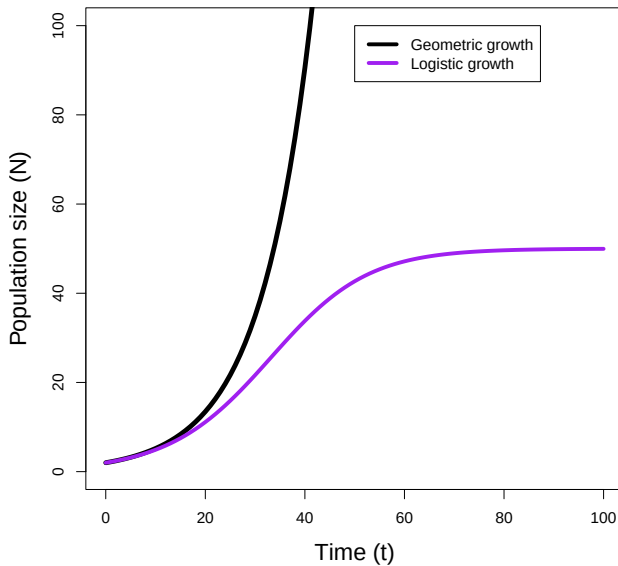
Definition: Population growth rate *is* affected by population size (N).

Implications: Resources are limited and there is a carrying capacity.

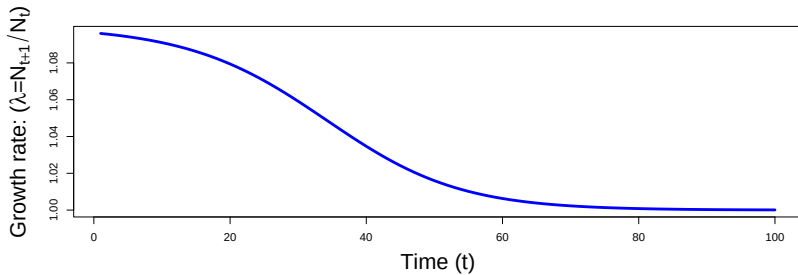
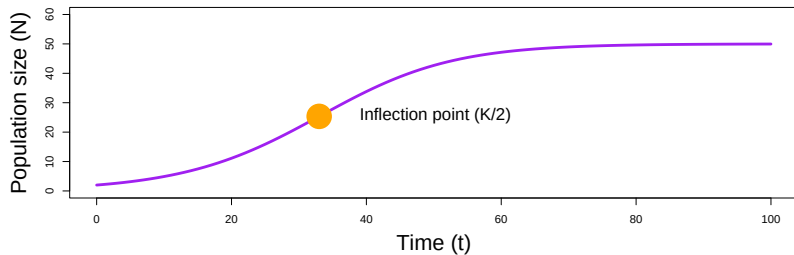
GRAPHICAL DEPICTION



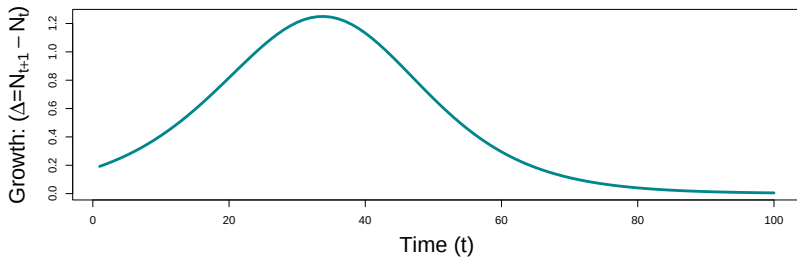
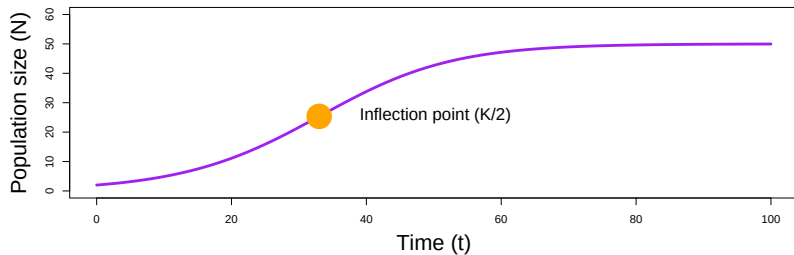
GRAPHICAL DEPICTION



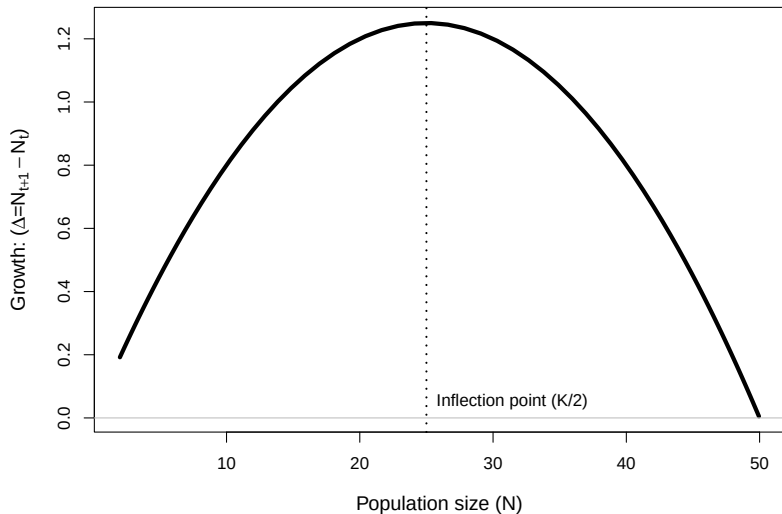
GROWTH RATE ($\lambda_t = N_{t+1}/N_t$)



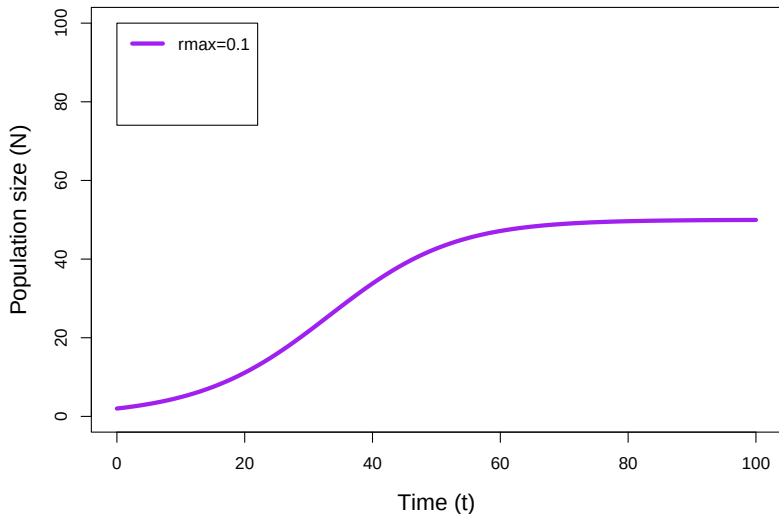
GROWTH ($\Delta_t = N_{t+1} - N_t$)



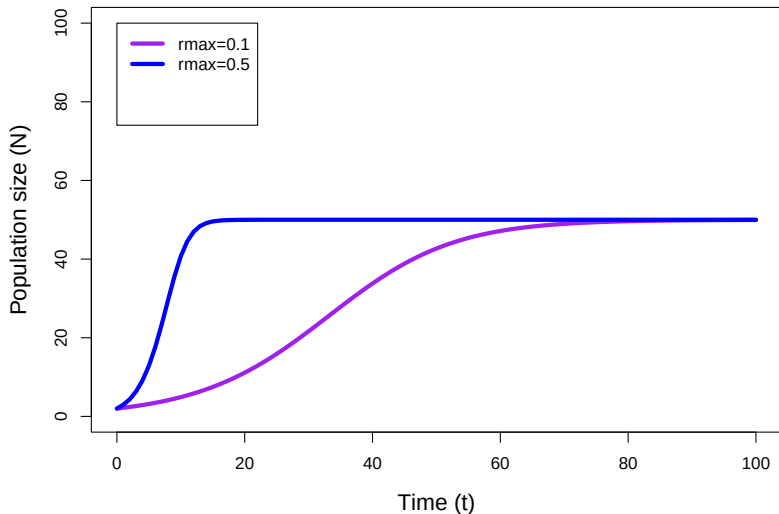
GROWTH ($\Delta_t = N_{t+1} - N_t$) AS A FUNCTION OF N



WHAT HAPPENS WHEN WE CHANGE r_{max} ?

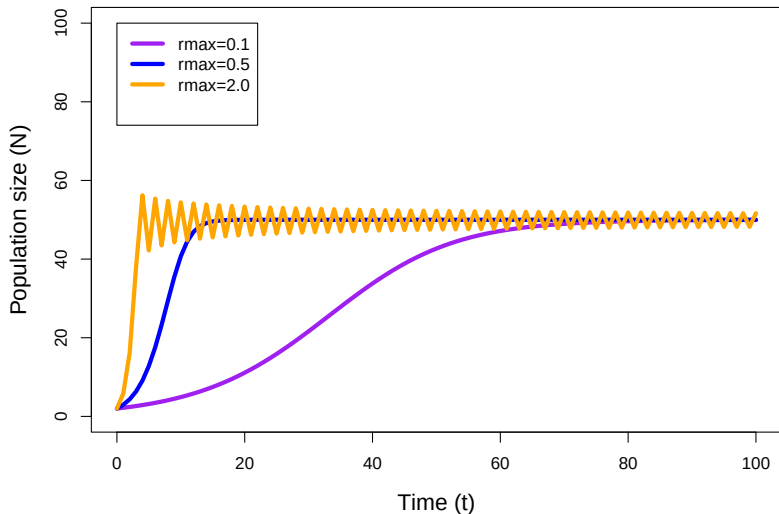


WHAT HAPPENS WHEN WE CHANGE r_{max} ?



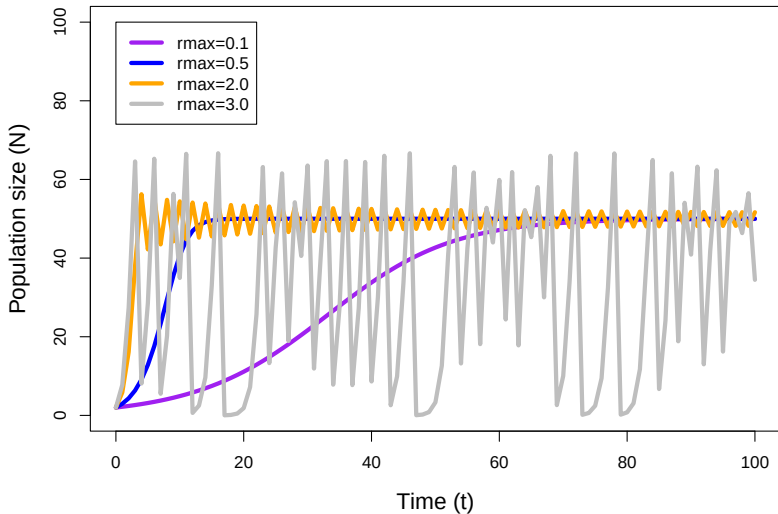
WHAT HAPPENS WHEN WE CHANGE r_{max} ?

Damped oscillation



WHAT HAPPENS WHEN WE CHANGE r_{max} ?

Chaos



OVERCOMPENSATION

Density-dependent response in which populations over- or under-shoot carrying capacity rather than approach it gradually

OVERCOMPENSATION

Density-dependent response in which populations over- or under-shoot carrying capacity rather than approach it gradually

CHAOS

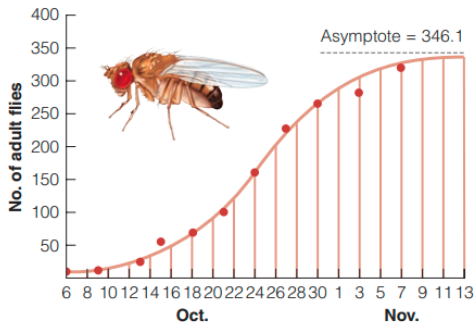
Highly variable deterministic dynamics that are extremely sensitive to small changes in parameters

- K and r_{max} are constant
- No sex or age effects or other sources of individual heterogeneity
- No time lags
- No stochasticity

DOES LOGISTIC GROWTH OCCUR IN NATURE?

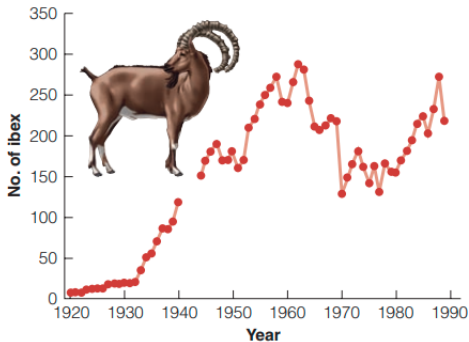
DOES LOGISTIC GROWTH OCCUR IN NATURE?

Fruit fly (*Drosophila melanogaster*) in the lab



DOES LOGISTIC GROWTH OCCUR IN NATURE?

Ibex (*Capra ibex*) in Switzerland.



Data from Sæther et al. (2002). Figure from Krebs (2014)

Summary

- Logistic growth is a form of density-dependent growth

Summary

- Logistic growth is a form of density-dependent growth
- Growth rate ($\lambda_t = N_{t+1}/N_t$) declines as N approaches K

Summary

- Logistic growth is a form of density-dependent growth
- Growth rate ($\lambda_t = N_{t+1}/N_t$) declines as N approaches K
- Growth ($\Delta_t = N_{t+1} - N_t$) peaks at $K/2$ (the inflection point)

Summary

- Logistic growth is a form of density-dependent growth
- Growth rate ($\lambda_t = N_{t+1}/N_t$) declines as N approaches K
- Growth ($\Delta_t = N_{t+1} - N_t$) peaks at $K/2$ (the inflection point)
- The model doesn't include birth, mortality, and movement processes, but it can be derived from density-dependent birth and death rates.

Summary

- Logistic growth is a form of density-dependent growth
- Growth rate ($\lambda_t = N_{t+1}/N_t$) declines as N approaches K
- Growth ($\Delta_t = N_{t+1} - N_t$) peaks at $K/2$ (the inflection point)
- The model doesn't include birth, mortality, and movement processes, but it can be derived from density-dependent birth and death rates.
- But it does allow for complex dynamics that resemble patterns seen in nature.

Summary

- Logistic growth is a form of density-dependent growth
- Growth rate ($\lambda_t = N_{t+1}/N_t$) declines as N approaches K
- Growth ($\Delta_t = N_{t+1} - N_t$) peaks at $K/2$ (the inflection point)
- The model doesn't include birth, mortality, and movement processes, but it can be derived from density-dependent birth and death rates.
- But it does allow for complex dynamics that resemble patterns seen in nature.

Assignment

Read pages 32–36 in Conroy and Carroll